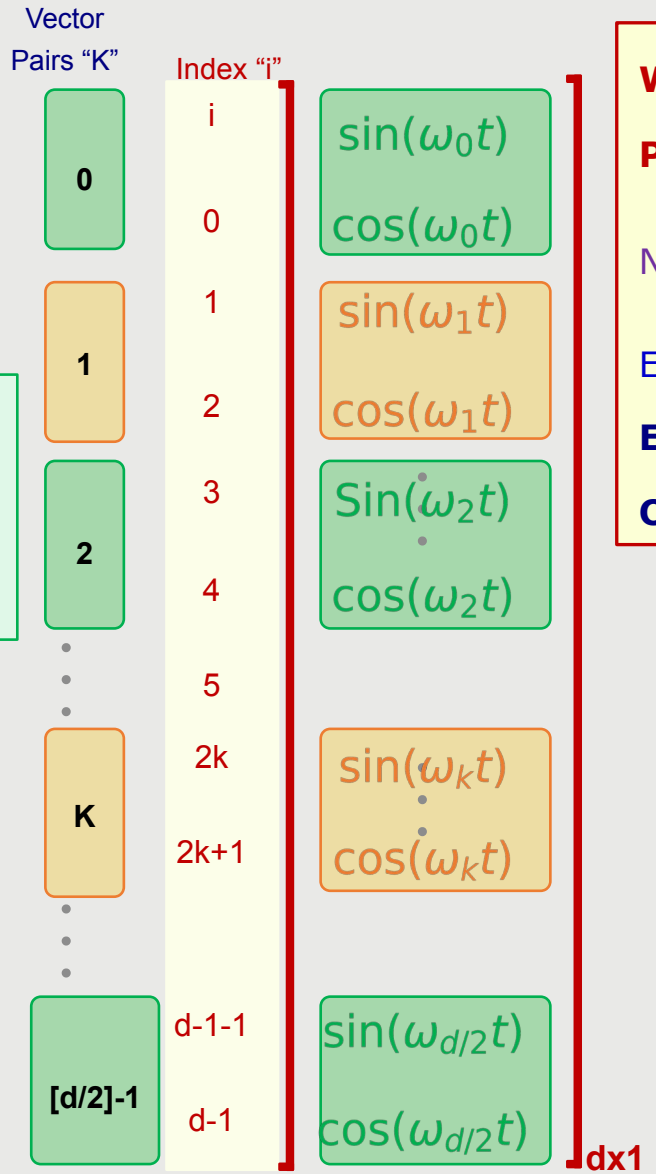
Sin/Cos Positional Encoder

Positional Vector with Same dimension as Embedding dimension

$$\begin{bmatrix} \sin(\omega t) \\ \cos(\omega t) \end{bmatrix}$$

Sin/Cos pairs
Vector Represents
Position "t"

Any "ω" is accepted



Word Embedding Dimension **"d"** ***[Even Num]***

Position Encoding Dimension **"d"**

Number of Vector Pairs "**K"=d/2** [0, 1, 2,... d/2-1]

Elements Index **"i"** [0, 1, 2, ..., d-1]

Even Index **"i"** is **Sine Term** , **i=2*k**

Odd Index **"i"** is **Cosine Term, i=2*k+1**

Example:

d=100

K=[0 ..49]

i=[0 .. 99]

Sin [0,2,4,...]

Cos [1,3,5,...]

Use different "ω" values for different "Pairs"
Select "ω" Value Based on "Pair" index k

$$\omega_K = \frac{1}{10{,}000^{2k/d}}$$

Largest "ω" value = 1 (Used in First Pair)
Smallest "ω" value ≈ 1/10000 (Used in Last Pair)

Note:
"ω" Value depends on Embedding Dimension "d" **ONLY**
"ω" Value does **NOT** depend on Sequence Length